Project - Low Level Design

On

Autonomous Energy Researcher Agent

Course Name: Agentic AI

Institution Name: Medicaps University – Datagami Skill Based Course

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1. Introduction

This document presents the Low-Level Design (LLD) of the Autonomous Energy Researcher Agent project. It provides a detailed technical description of the system by breaking down its internal structure, components, workflows, and interactions.

The purpose of this document is to define how the system is designed and implemented at a granular level. It includes detailed information about system modules, agent interactions, data flow, and execution processes.

This LLD serves as a blueprint for developers to implement the system and for evaluators to understand the design approach. It ensures clarity by translating high-level functionalities into detailed technical components.

1.1. Scope of the document

This document defines the detailed low-level design of the Autonomous Energy Researcher Agent. It describes:

- Internal architecture of the system
- Agent design and responsibilities
- Tool and API integration
- Data and process flow
- Execution workflow

The system focuses on automating energy-related research tasks, including data collection, analysis, and summarization using AI agents.

1.2. Intended Audience

This document is intended for:

- Developers implementing the system
- Academic evaluators and mentors
- Students studying Agentic AI
- Researchers exploring AI in the energy sector

It helps the audience understand the system design, workflows, and technical implementation.

1.3. System overview

The system is a **multi-agent AI-based research assistant** that processes user queries

Working Overview

- User provides an energy-related query
- Research Agent collects data from web sources
- Analysis Agent processes and filters data
- Summary Agent generates structured insights
- Final report is stored in the knowledge repository

Key Features

- Autonomous web-based research
- Multi-agent collaboration
- Intelligent summarization

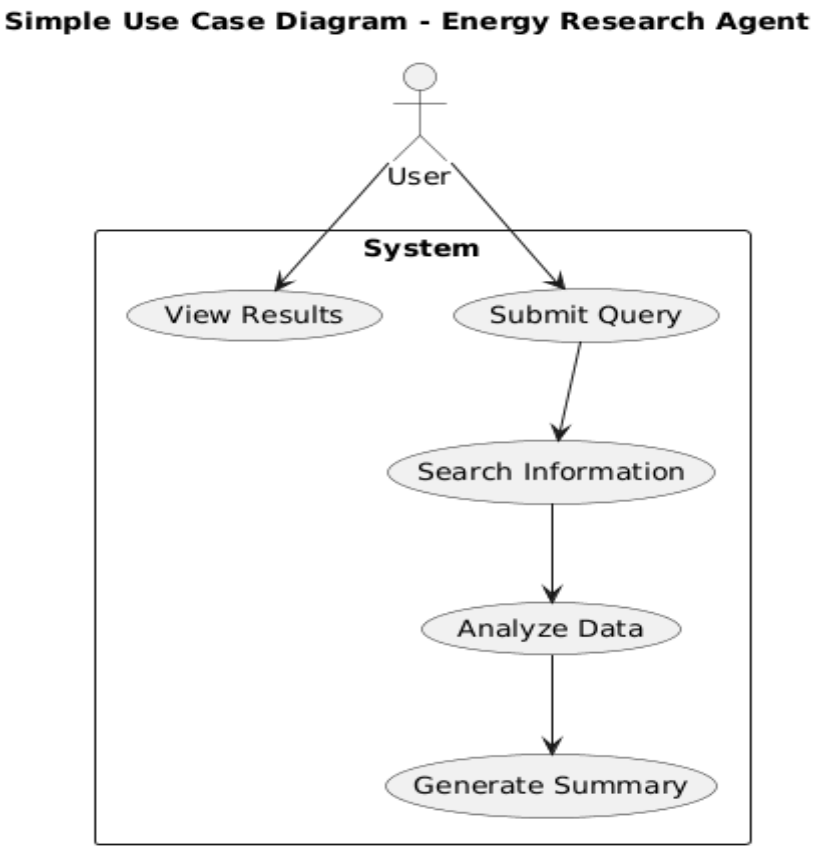

Figure 1: Use Case Diagram

2. System Design

The system follows a **modular multi-agent architecture** ensuring scalability, flexibility, and efficient execution.

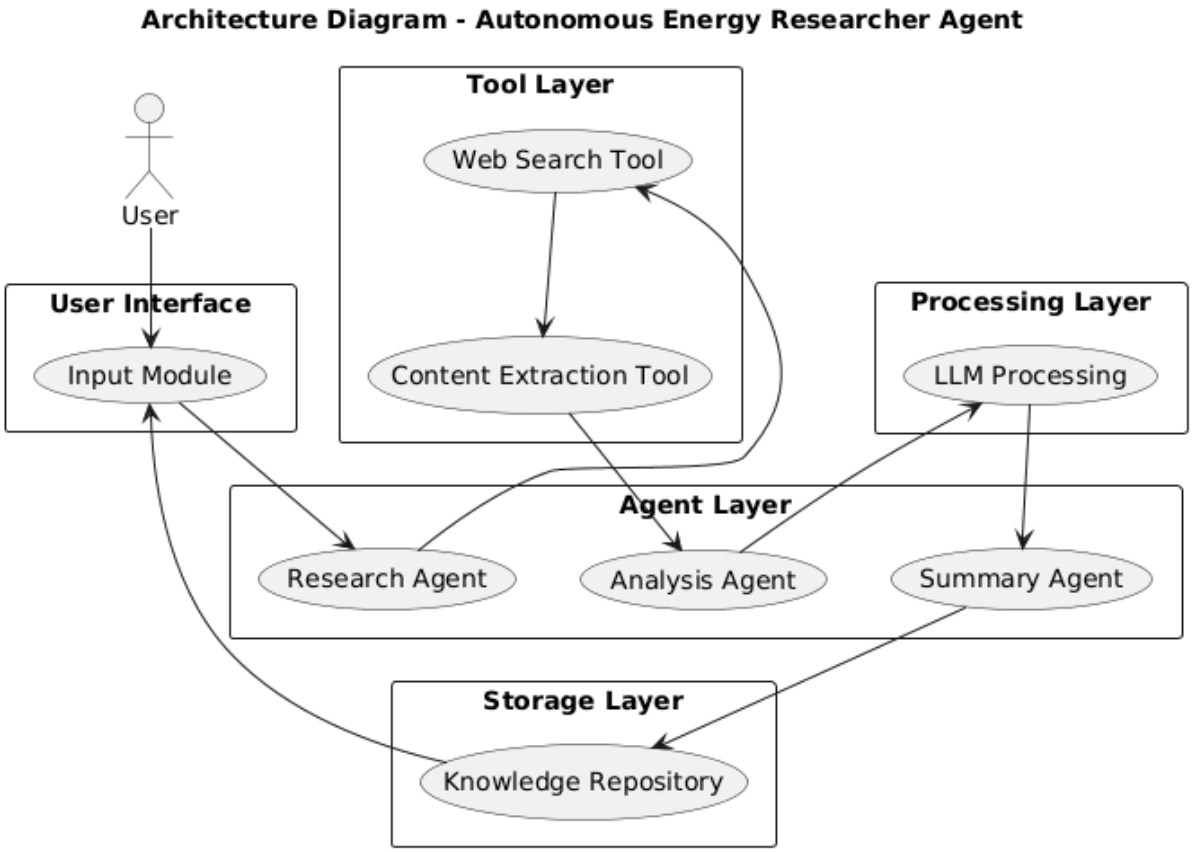

Figure 2: Architecture Diagram

Architecture Flow

User Input → Research Agent → Analysis Agent → Summary Agent → Knowledge Repository→Output

2.1. Application Design

The application consists of the following modules:

1. Input Module

- Accepts user query
- Validates input

- Preprocesses text

2. Research Agent

- Performs web search
- Collects relevant information
- Extracts content from sources

3. Agent Layer

Research Agent

- Collects raw data

Analysis Agent

- Filters and processes data
- Identifies key insights

Summary Agent

- Generates structured summaries
- Formats final output

4. Tool Layer

- Web Search Tool
- Content Extraction Tool
- Data Processing Utilities

5. Output Module

- Aggregates results
- Generates final research report
- Displays output to user

2.2. Process Flow

The system follows a structured execution flow:

1. User inputs energy-related query
2. System validates input
3. Research Agent performs web search
4. Data is collected from multiple sources
5. Analysis Agent filters and processes data
6. Summary Agent generates insights
7. Final report is created
8. Output is stored and displayed

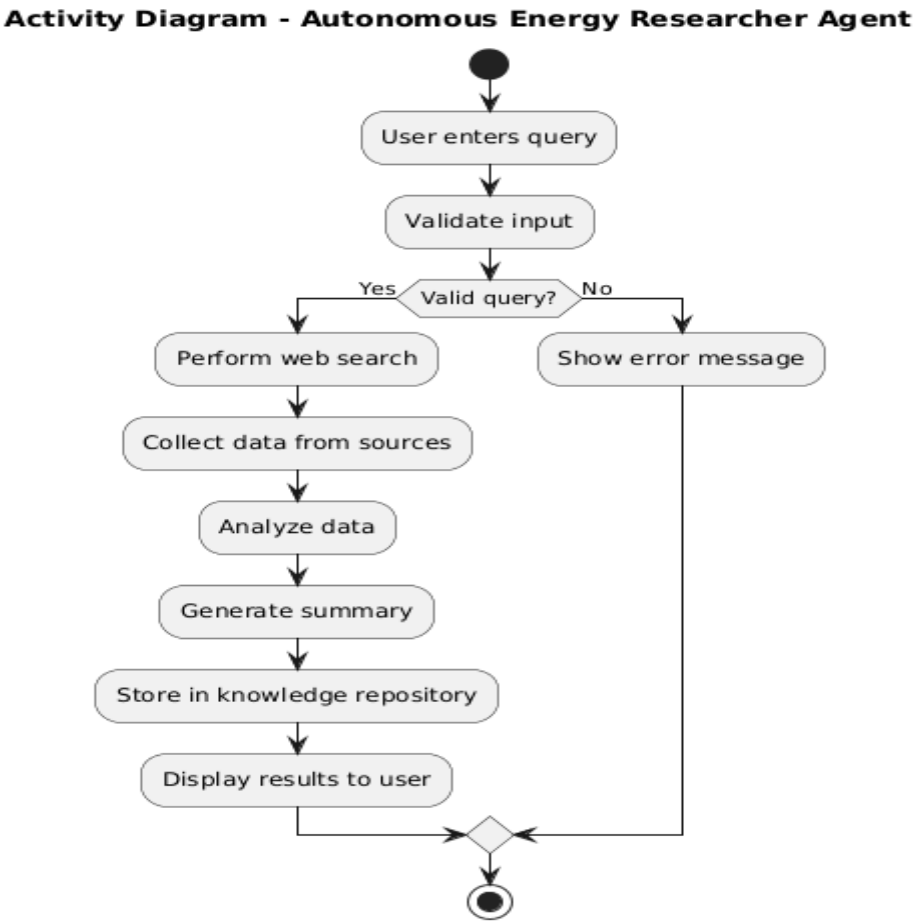

Figure 3: Activity Diagram

2.3. Information Flow

The flow of information in the system is as follows:

User Input
↓
Research Agent
↓
Raw Data
↓
Analysis Agents
↓
Processed Data
↓
Summary Agent
↓
Final Output

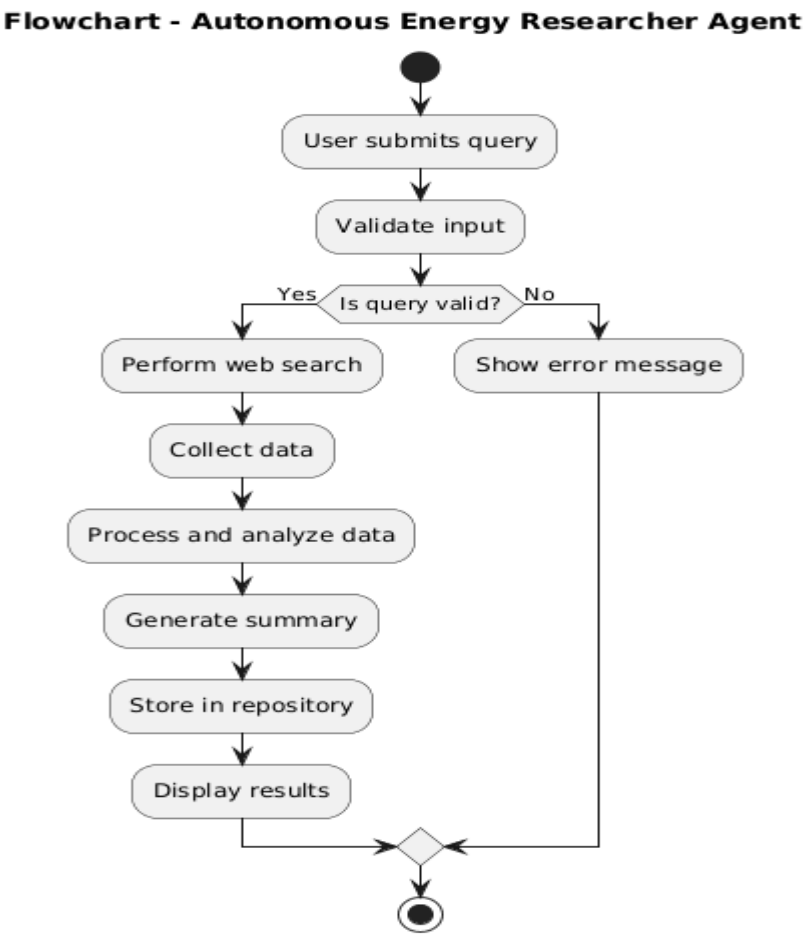

Figure 4: Flow Chart

Data Handling

- Input is cleaned and structured
- Data flows between agents
- Output is formatted for readability

2.4. Components Design

1. Research Agent
   - Performs web search
   - Extracts relevant content
2. Analysis Agent
   - Filters irrelevant data
   - Identifies key information
3. Summary Agent
   - Generates summarized report
   - Structures the output
4. Tool Layer
   - Provides web search functionality
   - Extracts and cleans content
5. Knowledge Repository
   - Stores research outputs
   - Maintains historical data
6. LLM Integration
   - Uses OpenAI API
   - Processes and generates insights

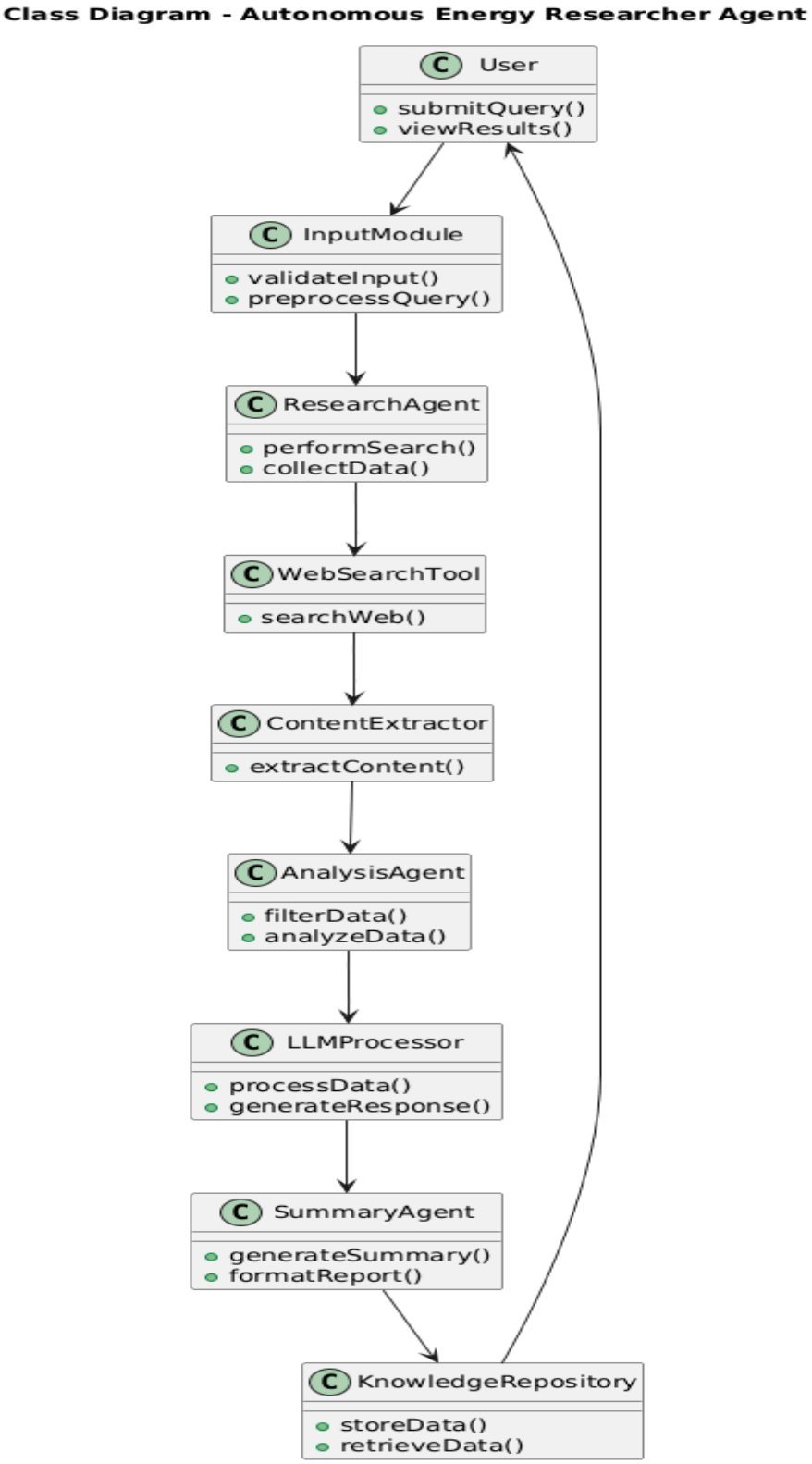

Figure 5: Class Diagram

2.5. Key Design Considerations

Scalability

The system can handle multiple agents and tasks simultaneously.

Modularity

Each component is independent and can be updated separately.

Flexibility

New agents and tools can be added easily.

Performance

Fast response using efficient LLM models.

Reliability

Mock tools ensure stable execution without external dependencies.

Security

No real patient data is used, ensuring data privacy.

2.6. API Catalogue

1. LLM API – OpenAI API

Parameter	Description
API Key	Authentication
Model	GPT-based model
Input	User query / processed data
Output	Generated response

2. Web Search API

Input: Query
Output: Search results, URLs

3. Content Extraction Tool

Input : Web URL
Output: Extracted text

4. Knowledge Storage API

Input : Research output
Output : Stored data

3. References

- LangChain Documentation
- CrewAI Documentation
- OpenAI API Documentation
- Research papers on Agentic AI
- Energy sector research resources